

Lecture 13: The Simplex Algorithm

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13.1 The Simplex Algorithm

Definition 13.1. *The standard form of a linear programming problem is*

$$\begin{aligned} &\text{minimize } z = \mathbf{c}^\top \mathbf{x} \text{ subject to} \\ &\mathbf{Ax} \leq \mathbf{b} \\ &\mathbf{x} \geq \mathbf{0}, \end{aligned}$$

where $\mathbf{A} = [a_{ij}]$ is an $m \times n$ matrix, $\mathbf{b} = [b_1 \ b_2 \ \cdots \ b_m]^\top$, $\mathbf{c} = [c_1 \ c_2 \ \cdots \ c_n]^\top$, and $\mathbf{x} = [x_1 \ x_2 \ \cdots \ x_n]^\top$

We need to make the assumption that $\mathbf{b} \geq \mathbf{0}$. We now turn the standard form into the canonical form, by adding in slack variables $x_{n+1}, \dots, x_{n+m}$. This means we have the LPP

$$\begin{aligned} &\text{minimize } z = \mathbf{c}^\top \mathbf{x} \text{ subject to} \\ &\mathbf{Ax} = \mathbf{b} \\ &\mathbf{x} \geq \mathbf{0}, \end{aligned}$$

where $\mathbf{A} = [a_{ij}]$ is the $m \times (n + m)$ matrix,

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & 1 & 0 & \cdots & 0 \\ a_{21} & a_{22} & \cdots & a_{2n} & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & 0 & 0 & \cdots & 1 \end{bmatrix}$$

and $\mathbf{c} = [c_1 \ c_2 \ \cdots \ c_n \ 0 \ \cdots \ 0]^\top$, and $\mathbf{x} = [x_1 \ x_2 \ \cdots \ x_n \ x_{n+1} \ \cdots \ x_{n+m}]^\top$

Definition 13.2. *Two distinct extreme points in S are said to be adjacent if as basic feasible solutions they have all but one basic variable in common.*

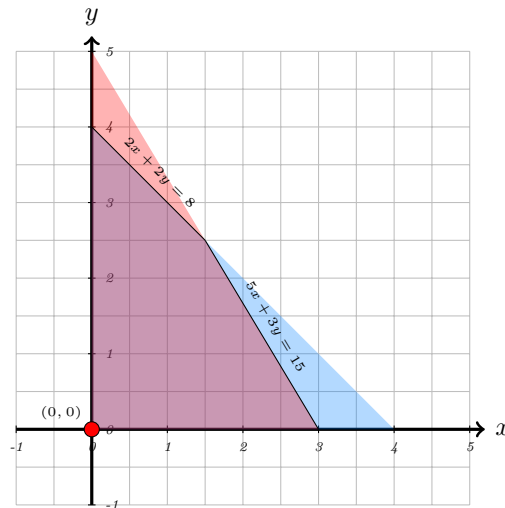
Example 13.3.

$$\begin{aligned} &\text{maximize } z = 120x + 100y \text{ subject to} \\ &2x + 2y \leq 8 \\ &5x + 3y \leq 15 \\ &x, y > 0 \end{aligned}$$

First, we need to convert to canonical form.

$$\begin{aligned} &\text{maximize } z = 120x + 100y \text{ subject to} \\ &2x + 2y + u = 8 \\ &5x + 3y + v = 15 \\ &x, y, u, v \geq 0 \end{aligned}$$

First, we send $x = y = 0$. This is possible because we still need 2 nonzero solutions, which in this case is $u = 8, v = 15$. This corresponds to



Lastly, we rewrite the objective function as $z - 120x - 100y = 0$. Now that we have all of the equations as equalities, we want to construct the initial tableau.

	x	y	u	v	z	b
u	2	2	1	0	0	8
v	5	3	0	1	0	15
	-120	-100	0	0	1	0

In the tableau, a basic variable has the following properties:

1. It appears in exactly one equation and in that equation it has a coefficient of +1
2. The column that it labels has all zeros (including the objective row entry) except for the +1 in the row that is labeled by the basic variable
3. The value of a basic variable is the entry in the same row in the rightmost column.

13.1.1 Optimality

If the objective row of a tableau has zero entries in the columns labeled by basic variables and no negative entries in the columns labeled by nonbasic variables, then the solution represented by the tableau is optimal.

13.1.2 Entering Variables

The simplex method proceeds from a given extreme point (basic feasible solution) to an adjacent extreme point in such a way that the objective function increases in value. From the definition of adjacent extreme point, it is clear that we reach such a point by increasing a single variable from zero to a positive value and decreasing a variable with a positive value to zero.

The largest increase in z per unit increase in a variable occurs for the most negative entry in the objective row. In Tableau 13.3, the most negative entry, -120, in the objective row occurs under the x column, so that x is chosen to be the variable to be increased from zero to a positive value.

13.1.3 Departing Variable

Since we are only allowed to have two basic variables, we need to select one to leave as well.

$$\begin{aligned}u &= 8 - 2x - 2y \\v &= 15 - 5x - 3y\end{aligned}$$

Since x is the entering variable, we increase that and keep y at zero.

$$\begin{aligned}u &= 8 - 2x \\v &= 15 - 5x\end{aligned}$$

As x increases, both u, v decrease. Remember, we can increase x until either u or v becomes negative, which means

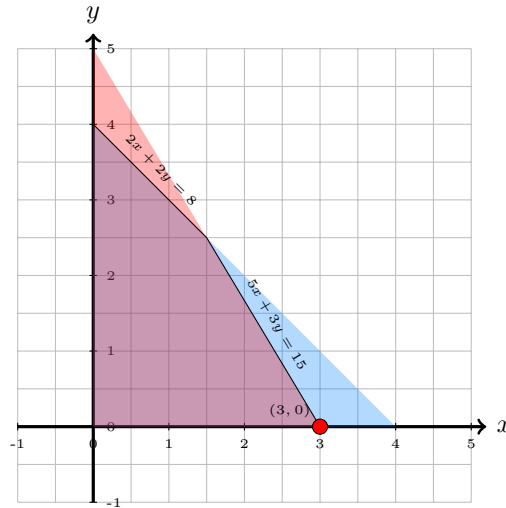
$$\begin{aligned}0 \leq u = 8 - 2x &\implies x \leq 4 \\- \leq v = 15 - 5x &\implies x \leq 3.\end{aligned}$$

This means we can only increase x by the smaller amount, which means $x = 3$, so we have a new feasible solution

$$x = 3, y = 0, u = 2, v = 0.$$

These two points are adjacent; $(0, 0, 8, 15) \rightarrow (3, 0, 2, 0)$, since x went from zero to nonzero, and v went from nonzero to zero (basic to nonbasic), or the middle two values didn't change. This means the objective function now has the value

$$z = 120 \cdot 3 + 100 \cdot 0 + 0 \cdot 2 + 0 \cdot 0 = 360.$$



In the new basic feasible solution to our example, we have the variable $v = 0$. It is no longer a basic variable because it is zero, and it is called a *departing variable* since it has departed from the set of basic variables. The column of the entering variable is called the *pivotal column*; the row that is labeled with the departing variable is called the *pivotal row*.

What if we picked the other ratio? We had $x \leq 8/2$ or $x \leq 15/5$. This means that the next basic solution is not feasible. If $x = 4$, we would have $u = 8 - 2 \cdot 4 = 0$, $v = 15 - 5 \cdot (4) = -5$, and we would have

$$x = 4, y = 0, u = 0, v = -5,$$

which is not feasible.

13.1.4 Forming a New Tableau

After we have figured out which have entered and departed, we need to make a new tableau. Solving the second constraint, we have

$$x = 3 - \frac{3}{5}y - \frac{1}{5}v,$$

which now means we have

$$2 \left(3 - \frac{3}{5}y - \frac{1}{5}v \right) + 2y + u = 8 \implies \frac{4}{5}y + u - \frac{2}{5}v = 2.$$

Substituting into the objective function, we have

$$(-120) \left(3 - \frac{3}{5}y - \frac{1}{5}v \right) - 100y + z = 0 \implies -28y + 24v + z = 360.$$

1. Locate and circle the entry at the intersection of the pivotal row and pivotal column. This entry is called the pivot. Mark the pivotal column by placing an arrow $\downarrow$ above the entering variable, and mark the pivotal row by placing an arrow $\leftarrow$ to the left of the departing variable.

	x	y	u	v	z	b
u	0	$\frac{4}{5}$	1	$-\frac{2}{5}$	0	2
x	1	$\frac{3}{5}$	0	$\frac{1}{5}$	0	3
	0	-28	0	24	1	360

2. If the pivot is k , multiply the pivotal row by $1/k$, making the entry in the pivot position in the new tableau equal to 1.
3. Add suitable multiples of the new pivotal row to all other rows (including the objective row), so that all other elements in the pivotal column become zero.
4. In the new tableau, replace the label on the pivotal row by the entering variable

Are we done? No because there is a zero under u but there is still a negative in the objective function row. We repeat.

Since -28 occurs in the y column that is the entering variable. To find the departing variables, we need to form the ratios. This means we compute

$$\frac{2}{\frac{4}{5}} = \frac{5}{2} \quad \text{and} \quad \frac{3}{\frac{3}{5}} = 5.$$

We find the lower of the two, which is $\frac{5}{2}$, so that means that u is the departing variable. Now we need to pivot.

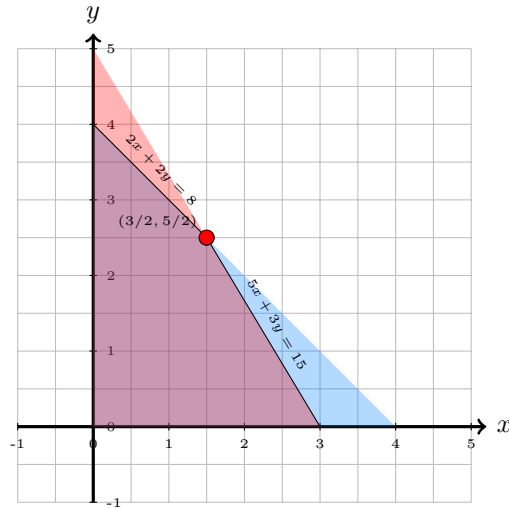
Since the pivot was $\frac{4}{5}$, we multiply everything by $\frac{5}{4}$ to get the pivotal row to be

	x	y	u	v	z	b
u	0	1	$\frac{5}{4}$	$-\frac{1}{2}$	0	$\frac{5}{2}$
x	1	$\frac{3}{5}$	0	$\frac{1}{5}$	0	3
	0	-28	0	24	1	360

Now we row reduce to get

	x	y	u	v	z	b
u	0	1	$\frac{5}{4}$	$-\frac{1}{2}$	0	$\frac{5}{2}$
x	1	0	$-\frac{3}{4}$	$\frac{1}{2}$	0	$\frac{3}{2}$
	0	0	35	10	1	430

Since there are no negative entries in the objective row, we are done, as we have met the optimality condition.

**Example 13.4.**

$$\begin{aligned}
 &\text{maximize } z = 4x_1 + x_2 + 4x_3 \text{ subject to} \\
 &2x_1 + x_2 + x_3 \leq 2 \\
 &x_1 + 2x_2 + 3x_3 \leq 4 \\
 &2x_1 + 2x_2 + x_3 \leq 8 \\
 &x_1, x_2, x_3 \geq 0
 \end{aligned}$$

First, we need to get it to canonical form

$$\begin{aligned}
 z - 4x_1 - x_2 - 4x_3 &= 0 \\
 2x_1 + x_2 + x_3 + u &= 2 \\
 x_1 + 2x_2 + 3x_3 + v &= 4 \\
 2x_1 + 2x_2 + x_3 + w &= 8 \\
 x_1, x_2, x_3, u, v, w &\geq 0
 \end{aligned}$$

Now we need to set up the initial tableau:

	x_1	x_2	x_3	u	v	w	z	b
u	2	1	1	1	0	0	0	2
v	1	2	3	0	1	0	0	4
w	2	2	1	0	0	1	0	8
	-4	-1	-4	0	0	0	1	0

Are there negative entries in the objective row? YES! Are there any positive entries in the pivotal column above the objective row? YES! Compute the new tableau by pivoting.

We need to compute the pivot. The first row we have, $\frac{1}{2} = 1$ and the second row we have $\frac{4}{1} = 4$. Since we need to pick the smaller of the two $\min 1, 4 = 1$, so the pivot is x_1 and u will be departing u .

Now we row reduce to get the new pivot to be a basic variable

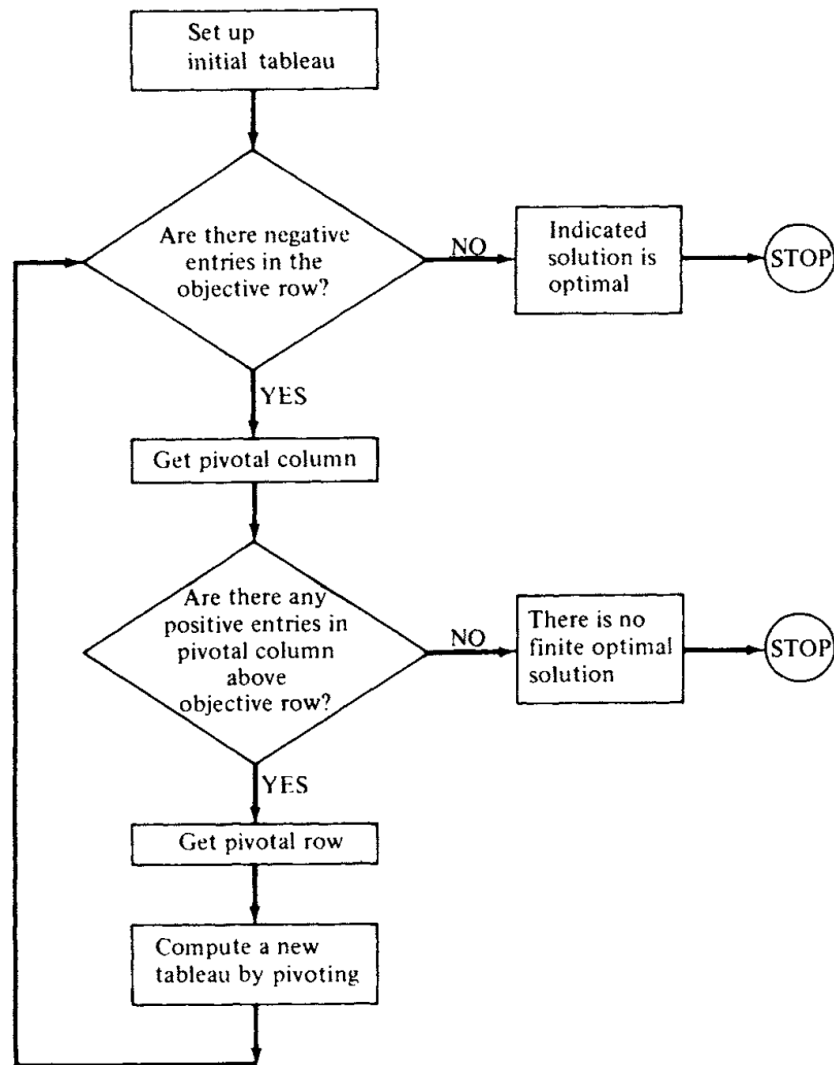


Figure 13.1:

	x_1	x_2	x_3	u	v	w	z	b
u	1	0.5	0.5	0.5	0	0	0	1
v	1	2	3	0	1	0	0	4
w	2	2	1	0	0	1	0	8
	-4	-1	-4	0	0	0	1	0

	x_1	x_2	x_3	u	v	w	z	b
x_1	1	0.5	0.5	0.5	0	0	0	1
v	0	1.5	2.5	-0.5	1	0	0	3
w	0	1	0	-1	0	1	0	6
	0	1	-2	2	0	0	1	4

Are there negative entries in the objective row? YES! Are there any positive entries in the pivotal column above the objective row? YES! Compute the new tableau by pivoting.

We need to compute the pivot. The first row we have, $\frac{1}{0.5} = 2$ and the second row we have $\frac{3}{2.5} = \frac{6}{5} = 1.2$. Since we need to pick the smaller of the two $\min 2, 1.2 = 1.2$, so the pivot is x_2 and v will be departing us.

	x_1	x_2	x_3	u	v	w	z	b
x_1	1	0.5	0.5	0.5	0	0	0	1
x_3	0	0.6	1	-0.2	0.4	0	0	1.2
w	0	1	0	-1	0	1	0	6
	0	1	-2	2	0	0	1	4

	x_1	x_2	x_3	u	v	w	z	b
x_1	1	0.2	0	-1.2	0.6	0	0	0.4
x_3	0	0.6	1	-0.2	0.4	0	0	1.2
w	0	1	0	-1	0	1	0	6
	0	2.2	0	1.6	0.8	0	1	6.4

Are there negative entries in the objective row? NO! We have met the optimality condition.

Example 13.5.

maximize $z = 2x + 5y$ subject to

$$3x + 5y \leq 8$$

$$2x + 7y \leq 12$$

$$x, y \geq 0$$